Observer-Patch Holography Cosmology Data and Likelihood Contracts

Provenance, Comparability, and Claim Boundaries

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Abstract

We specify data and likelihood contracts for conditional Observer-Patch Holography (OPH) cosmology. Each comparison binds a finite source, compatible transfer equations, declared datasets and covariance, nuisance parameters, and a common observable projection. The contracts distinguish source-derived quantities, imported inputs and fitted parameters, and account for sample overlap and shared calibration. They give a reproducible interface from screen and dark-sector models to cosmological data.

1 Comparison Classes

Every reported number belongs to one of four classes:

1. **source-derived**: all physical inputs follow from the declared OPH construction without using the comparison data;
2. **imported**: established background or microphysical inputs are declared explicitly;
3. **fit**: one or more parameters are inferred from the same data;
4. **diagnostic**: the calculation tests shape, scale, code behavior, or an interface without supporting a physical OPH claim.

The cosmology comparisons in scope are diagnostic or imported. No source-derived joint cosmology likelihood exists.

The spacetime construction provides a flat causal-order and normalized count-volume limit for conservative record populations with a specified neighbour-read law and layer duration. With complete authenticated ancestry and anchors, the interval-count ratio $(|I|/|J|)^{1/4}$ approaches the proper-time ratio of interior timelike diamonds, relative to a nondegenerate reference interval in the same history. The readout needs no coordinates, timestamps or density; its convergence uses the supplied population and law.

A separate cone-selection criterion requires a nonzero closed convex pointed cone invariant under the proper icosahedral rotations and all rapidities of one boost axis. Its operational action on histories, interventions and clock readouts is additional to the algebraic source frame. In a likelihood comparison, that hypothesis, the event law and its physical calibration count as model information, alongside the background, conserved stress, initial conditions and matter assumptions. A geometric count clock by itself does not define the cosmological transfer or observable likelihood.

2 Required Model Information

A physical comparison must state:

1. the covariant action and source map;
2. the background solution and species content;
3. the gauge-invariant initial conditions;
4. the perturbation, collision, and exchange equations;
5. the observable projection and numerical tolerances;
6. every external input and fitted parameter.

For the modular-charge dark-sector proposal, conserved charge in a supplied comoving volume gives the homogeneous dilution law $\rho_A \propto a^{-3}$. With the separate no-exchange continuity equation it gives a pressureless background. The abundance, relativistic stress, initial perturbations, lensing map, clusters, and nonlinear structure equations are not supplied here.

3 Required Data Information

A comparison must identify the data release, selection, masks, covariance, calibration model, nuisance parameters, priors, and combination rule. Dataset overlap and shared calibration must be included. A diagonal sum is not a joint likelihood when the covariance is non-diagonal.

The model and data layers are separate. Measured values that select a source formula, normalization, branch, or applicability domain are inputs to that comparison rather than OPH outputs.

4 Observable Contracts

Observable family	Required theory output	Required comparison object
Cosmic microwave background (CMB)	C_ℓ^{TT}, C_ℓ^{TE}, C_ℓ^{EE}, lensing, foreground model	map or band-power likelihood with covariance and nuisance model
Baryon acoustic oscillations (BAO) and supernovae	$H(z)$, $D_A(z)$, luminosity distance, sound horizon	survey likelihood with calibration covariance
Weak lensing	metric potentials, nonlinear matter power, intrinsic alignments	tomographic likelihood with masks and covariance
Growth and redshift-space distortions (RSD)	$P_m(k,z)$ and $f\sigma_8(z)$	windowed survey likelihood with cross-bin covariance
Clusters	mass function, selection, observable–mass map, lensing calibration	count and calibration likelihood
Galaxies	disk dynamics, gas and stellar nuisances, external environment	galaxy-level likelihood with selection and covariance

5 Diagnostic Results

The reported CMB overlays, compressed background comparisons, and galaxy scaling checks are diagnostic. Their formulas or inputs are selected with knowledge of the comparison data, omit required covariance or nuisance structure, or lack a physical source map. They test numerical paths and conditional formulas and do not establish a cosmological prediction.

6 No-Data-Use Boundary

A source-derived calculation must expose its dependency graph. The source side may use mathematical constants, declared OPH finite artifacts, and explicitly imported physical inputs. It may not use the observed quantity being claimed as an output, a fit to that quantity, or a proxy calibrated from the same dataset.

This provenance rule is enforced through the declared dependency graph and data-use classification.

7 Theorem scope

The screen-spectrum and radial-lift mathematics are conditional theorems. One-shell inversion is nonunique; physical source dilation and radial cross-covariance tomography are two established sufficient uniqueness routes. Finite source instantiation, repair-charge cosmological source, Boltzmann bridge, and joint likelihood are not supplied. All cosmological data products are diagnostic. No cosmological result carries an OPH falsification verdict.

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